

# Math 133 Written Homework 2 Solutions

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0. **Assignment record.** Answers depend on the student's collaborators, time spent, and use of generative AI.

1. **A shipping fee that jumps.**

(a) The first rule applies at four pounds; the second applies at five pounds:

$$S(4) = 4 + 4 = 8, \quad S(5) = 9 + \frac{3}{4}(5) = 12.75.$$

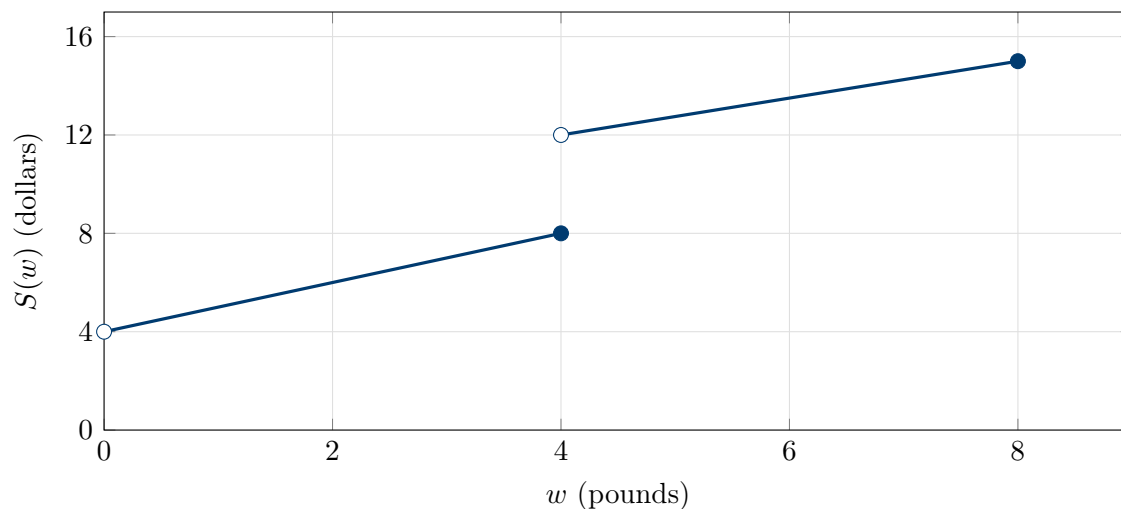
The graph has a segment from  $(0, 4)$  (open) to  $(4, 8)$  (filled), and a segment from  $(4, 12)$  (open) to  $(8, 15)$  (filled).

(b) The three differences are

$$S(5) - S(4) = 12.75 - 8 = 4.75, \quad S(2) - S(1) = 6 - 5 = 1,$$

$$S(6) - S(5) = 13.50 - 12.75 = 0.75.$$

Shipping a five-pound package costs \$4.75 more than shipping a four-pound package. Shipping a two-pound package costs \$1 more than shipping a one-pound package. Shipping a six-pound package costs \$0.75 more than shipping a five-pound package.

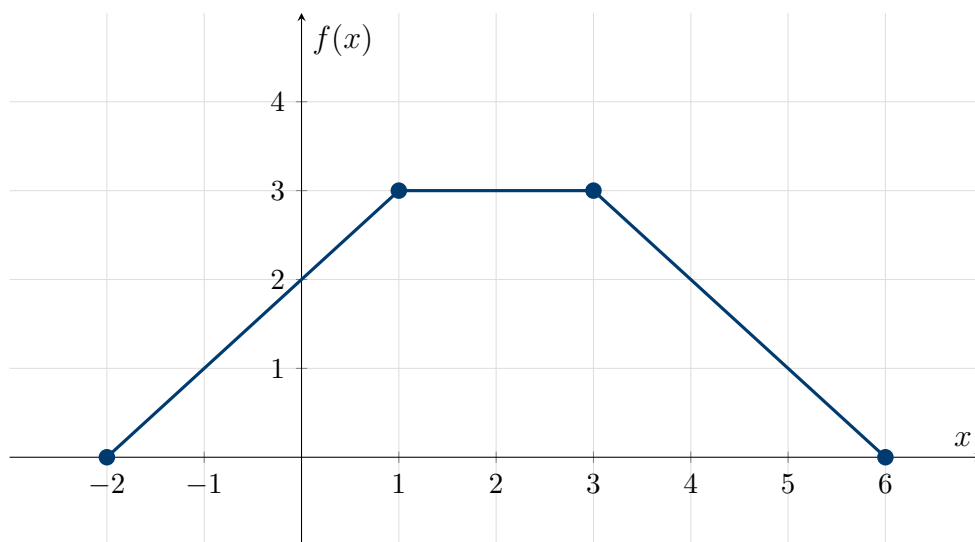


## 2. Graphing one piece at a time.

(a) Substituting into the applicable rule gives

| Input | Calculation          | Rule used |
|-------|----------------------|-----------|
| -2    | $f(-2) = -2 + 2 = 0$ | $x + 2$   |
| 0     | $f(0) = 0 + 2 = 2$   | $x + 2$   |
| 1     | $f(1) = 1 + 2 = 3$   | $x + 2$   |
| 2     | $f(2) = 3$           | 3         |
| 3     | $f(3) = 3$           | 3         |
| 6     | $f(6) = 6 - 6 = 0$   | $6 - x$   |

(b) Join  $(-2, 0)$  to  $(1, 3)$ , then  $(1, 3)$  to  $(3, 3)$ , then  $(3, 3)$  to  $(6, 0)$ . All four points belong to the graph. At each joining point, one rule includes the point, so the combined graph has a filled dot.

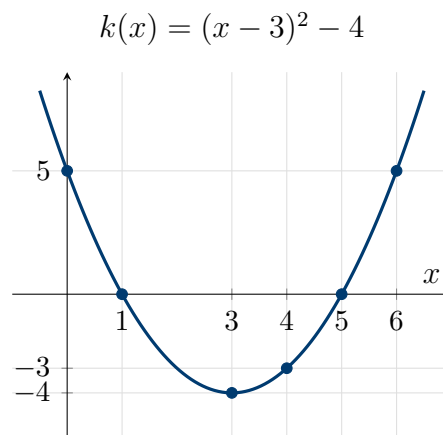
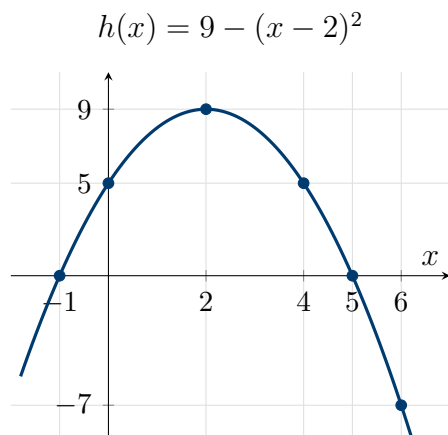


(c) Using the requested open-interval convention,  $f$  is increasing on  $(-2, 1)$ , constant on  $(1, 3)$ , and decreasing on  $(3, 6)$ . Its range is  $[0, 3]$ . The maximum value is 3, attained at every  $x \in [1, 3]$ , including both endpoints of the horizontal segment.

### 3. Quadratic graphs and restricted intervals.

- (a) For  $h(x) = 9 - (x - 2)^2$ , the vertex is  $(2, 9)$  and the parabola opens downward. Solving  $(x - 2)^2 = 9$  gives intercepts  $(-1, 0)$  and  $(5, 0)$ ;  $h(0) = 5$  gives  $(0, 5)$ .

For  $k(x) = x^2 - 6x + 5 = (x - 3)^2 - 4$ , the vertex is  $(3, -4)$  and the parabola opens upward. Factoring  $k(x) = (x - 1)(x - 5)$  gives intercepts  $(1, 0)$  and  $(5, 0)$ ;  $k(0) = 5$  gives  $(0, 5)$ .



- (b)  $h$  increases on  $(-\infty, 2)$  and decreases on  $(2, \infty)$ . The function  $k$  decreases on  $(-\infty, 3)$  and increases on  $(3, \infty)$ .
- (c) On  $[0, 4]$ , both vertices are included. Compare  $h(0) = h(4) = 5$  with  $h(2) = 9$ , and  $k(0) = 5$ ,  $k(4) = -3$  with  $k(3) = -4$ :

| Function | Maximum on $[0, 4]$ | Minimum on $[0, 4]$      |
|----------|---------------------|--------------------------|
| $h$      | 9 at $x = 2$        | 5 at $x = 0$ and $x = 4$ |
| $k$      | 5 at $x = 0$        | -4 at $x = 3$            |

- (d) On  $[4, 6]$ , neither vertex is included. The function  $h$  decreases throughout this interval, while  $k$  increases. The extrema therefore occur at the endpoints:

| Function | Maximum on $[4, 6]$ | Minimum on $[4, 6]$ |
|----------|---------------------|---------------------|
| $h$      | 5 at $x = 4$        | -7 at $x = 6$       |
| $k$      | 5 at $x = 6$        | -3 at $x = 4$       |

#### 4. A nonlinear demand model.

- (a) The price values are  $p(0) = 60$ ,  $p(2) = 30$ ,  $p(4) = 20$ , and  $p(10) = 10$  dollars per kilogram. As  $q$  increases, the positive denominator  $q + 2$  increases while the numerator stays fixed, so price decreases on  $[0, 10]$ .

- (b) Revenue is

$$R(q) = \frac{120q}{q+2}, \quad R(0) = 0, \quad R(2) = 60, \quad R(4) = 80, \quad R(10) = 100$$

in dollars. A decreasing price need not mean decreasing revenue: these revenues rise while the prices fall. More kilograms are sold at the lower prices. Indeed,  $R(q) = 120 - 240/(q+2)$  increases throughout this domain.

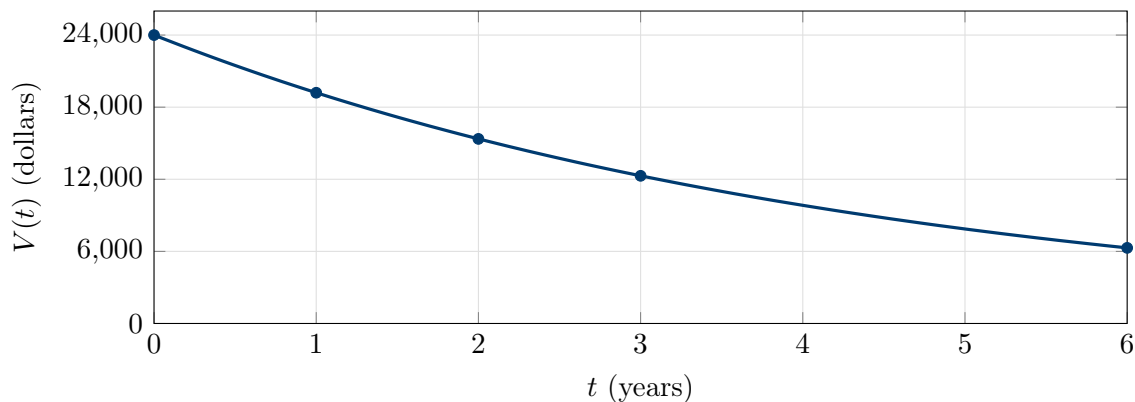
#### 5. An exponential depreciation model.

- (a) The initial value is \$24,000. The factor 0.80 means the van retains 80% of its value each year, so it loses 20% of its current value per year.

- (b) The requested values, rounded to the nearest dollar, are

| $t$ (years)      | 0     | 1     | 2     | 3     | 6    |
|------------------|-------|-------|-------|-------|------|
| $V(t)$ (dollars) | 24000 | 19200 | 15360 | 12288 | 6291 |

The model decreases on  $[0, 6]$ . Its maximum is \$24,000 at  $t = 0$ ; its minimum is  $24000(0.8)^6 = \$6,291.456$ , or \$6,291 to the nearest dollar, at  $t = 6$ .



- (c) The first-year loss is  $24000 - 19200 = \$4,800$ ; the second-year loss is  $19200 - 15360 = \$3,840$ . The dollar losses are not constant. Each year's value is multiplied by 0.80, so the model is exponential; a linear model would have a constant dollar change per year.

## 6. Excel investigation: choosing an advertising budget.

- (a) With budgets in A2:A12, enter the following in B2 and fill down:

$$=1800*(1-EXP(-0.4*A2))-100*A2-15*A2^2-200$$

The left table below gives the whole-number inputs. The right table gives the refined inputs from part (c). Values are rounded only for display.

- (b) The whole-interval graph has its highest plotted point at  $x = 3$ , with  $P(3) \approx \$622.85$ . This gives an initial budget estimate of \$300. Neither endpoint maximizes profit:  $P(0) = -\$200$  and  $P(10) \approx -\$932.97$ , both below  $P(3)$ .
- (c) Use inputs from 2 through 4 in increments of 0.1. The largest sampled profit is  $P(3.2) \approx 625.932859$ , so the refined advertising budget is \$320, with predicted profit \$625.93. Nearby values are  $P(3.1) \approx \$624.96$  and  $P(3.3) \approx \$625.81$ .
- (d) A table and connected-marker graph sample finitely many budgets; a higher value may occur between samples. Thus  $x = 3.2$  is an estimate of the maximizing input. At the resolution shown, profit peaks near this amount and then falls as the budget increases further.

Whole-number inputs

| $x$ | $P(x)$ (dollars) |
|-----|------------------|
| 0   | -200.00          |
| 1   | 278.42           |
| 2   | 531.21           |
| 3   | 622.85           |
| 4   | 596.59           |
| 5   | 481.40           |
| 6   | 296.71           |
| 7   | 55.54            |
| 8   | -233.37          |
| 9   | -564.18          |
| 10  | -932.97          |

Refined inputs

| $x$ | $P(x)$ (dollars) |
|-----|------------------|
| 2   | 531.21           |
| 2.1 | 546.77           |
| 2.2 | 560.79           |
| 2.3 | 573.32           |
| 2.4 | 584.39           |
| 2.5 | 594.07           |
| 2.6 | 602.38           |
| 2.7 | 609.38           |
| 2.8 | 615.10           |
| 2.9 | 619.57           |
| 3   | 622.85           |
| 3.1 | 624.96           |
| 3.2 | 625.93           |
| 3.3 | 625.81           |
| 3.4 | 624.61           |
| 3.5 | 622.38           |
| 3.6 | 619.13           |
| 3.7 | 614.90           |
| 3.8 | 609.72           |
| 3.9 | 603.61           |
| 4   | 596.59           |

**Problem 6: graphs of the tabulated values.**

The graphs below connect the same sampled values as the requested Excel scatterplots.

